

## **DSS and Expert System**

**Course Title:** DSS and Expert System  
**Course No:** BIT405  
**Nature of the Course:** Theory + Lab  
**Semester:** VII

**Full Marks:** 60+ 20+20  
**Pass Marks:** 24+8+8  
**Credit Hrs:** 3

### **Course Description:**

This course is a study uses of artificial intelligence in business decision making. Emphasis will be given in business decision making process, design and development of decision support systems and expert systems.

### **Course Objectives:**

To introduce intelligent business decision making, to describe design, development and evaluation of DSS Systems, to know various models of building DSS systems and to explain concept behind expert systems

### **Course Contents:**

#### **Unit 1: Introduction to management Support Systems and Decision making (14 Hrs.)**

Managers and decision making: Bounded rationality, muddling through; Factors in decision making: memory, bias, intuition, experience, models, analytics; Qualitative vs. quantitative decision making; Managerial decision making and information systems; Computerized decision support and supporting technologies; Group decision making, groupware Supporting Business Decision Making: Introduction, History, Conceptual Perspective, Decision Support vs. Transaction Processing System, Categories of DSS Applications and Products, DSS Framework, Building Decision Support Systems; Gaining Competitive Advantage with Decision Support Systems: Introduction, Technology Trends, Gaining Competitive Advantage, Examples of Strategic DSS, Opportunities and IS Planning, DSS Benefits, Limitations, and Risks, Resistances to Using DSS

#### **Unit 2: The Make-up of a Decision Support System (15 Hrs.)**

Types and roles of DSS. Data component: data vs. information, quality of information, databases and database management systems, data warehouses; Model component: representation, linearity of the relationship, deterministic vs. Stochastic, descriptive vs. Normative, causality vs. Correlation, methodology dimension, data mining and intelligent agents; Knowledge Engine Component; User Interface: Action language, display or presentation language, interface issues; User, Designing and Evaluating DSS Systems: Introduction, Design and Development Issues, Decision Oriented Diagnosis, Prepare a Feasibility Study, Choose a Development Approach, DSS Project Management and Participants; Designing and Evaluating DSS User Interfaces: Introduction, Overview of User Interface, User Interface Styles, ROMC Design Approach, Building DSS User Interface, Comments on Design Elements, Guidelines of Dialog and UI Design, Factors of UI Design Success

**Unit 3: Modeling Decisions (6 Hrs.)**

Introduction to Decision Analysis; Elements of Decision Problems, uncertain events, consequences, Structuring Decisions, Making Choices, making decisions with multiple objectives, assessing trade-of weights, Sensitivity Analysis, value of Information and experts

**Unit 4: Expert Systems (10 Hrs.)**

Definition and Features of Expert Systems, Architecture and Components of Expert Systems, Persons Who Interact with Expert Systems, Advantages and Disadvantages of Expert Systems, Expert Systems Development Life Cycle, Error Sources on Expert System Development

**Laboratory Works:**

Student should study and develop decision support systems or expert systems as a mini-project.

**References:**

1. Daniel J. Power, Decision Support Systems: Concepts and Resources for Managers, Illustrated Edition, Praeger.
2. I. Gupta and G. Nagpal, Artificial Intelligence and Expert Systems, Mercury Learning & Information, 2020